Homework #4

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1.)

$$P_0 y'' + P_1 y' + P_2 y = -\tilde{\lambda} y \quad (1)$$

$$(Py')' + qy = -\lambda w y \quad (2)$$

a.)

- multiply w to (1)

$$wP_0 y'' + wP_1 y' + wP_2 y = -w\tilde{\lambda} y$$

- <u>by inspection</u> -

$P = wP_0$, $P' = wP_1$, $q = wP_2$, $\lambda = \tilde{\lambda}$

$$\underbrace{Py'' + P'y'}_{\text{(product Rule)}} + qy = -\lambda w y$$

$$\boxed{(Py')' + qy = -\lambda w y}$$

b.)

$$\rho = \exp\left(\int \frac{\rho_1}{\rho_0} dx\right)$$

- $\rho = w\rho_0$, $\rho' = w\rho_1$
- $\frac{\rho}{\rho_0} = w = \frac{\rho'}{\rho_1}$
- $\frac{\rho'}{\rho_1} = \frac{\rho}{\rho_0}$
- $\frac{\rho'}{\rho} = \frac{\rho_1}{\rho_0}$
- $\int \frac{d\rho}{\rho} = \int \frac{\rho_1}{\rho_0} dx$
- $\ln(\rho) = \int \frac{\rho_1}{\rho_0} dx$
- $\boxed{\rho = \exp\left(\int \frac{\rho_1}{\rho_0} dx\right)}$

c.) $\exp\left(\int \frac{\rho_1}{\rho_0} dx\right) = w\rho_0$

$$\boxed{w = \frac{1}{\rho_0} \exp\left(\int \frac{\rho_1}{\rho_0} dx\right)}$$

2.)

a.) $y'' + (1-x)y' + \lambda y = 0$

(Laguerre ODE)

$(py')' + qy = -\lambda w y$

(Sturm-Liouville ODE)

$$w = \frac{1}{p_0} \exp\left(\int \frac{p_1}{p_0} dx\right)$$

$$p_0 y'' + p_1 y' + p_2 y = -\tilde{\lambda} y$$

- $p_0 = 1$
- $p_1 = (1-x)$
- $p_2 = \lambda$

$$w = \exp\left(\int (1-x) dx\right)$$

$$\int (1-x) dx = x - \frac{x^2}{2} + C$$

$$w = \exp\left(x - \frac{x^2}{2} + C\right)$$

$$W = \exp\left(x\left(1-\frac{x}{2}\right) + C\right)$$

$$A = \exp(C)$$

$$\boxed{W = A\exp\left(x\left(1-\frac{x}{2}\right)\right)}$$

$$A\exp\left(x\left(1-\frac{x}{2}\right)\right)y'' + A(1-x)\exp\left(x\left(1-\frac{x}{2}\right)\right)y'$$

$$+ \lambda\exp\left(x\left(1-\frac{x}{2}\right)\right)y = 0$$

$$\boxed{\frac{d}{dx}\left(A\exp\left(x\left(1-\frac{x}{2}\right)\right)y'\right) = -\lambda\exp\left(x\left(1-\frac{x}{2}\right)\right)y}$$

b.)

$$y'' - 2xy' + 2\lambda y = 0$$

(Hermite ODE)

$$W = \frac{1}{P_0} \exp\left(\int \frac{P_1}{P_0} dx\right)$$

$P_0 = 1, \quad P_1 = -2x$

$$W = \exp\left(\int -2x\,dx\right)$$

$$W = \exp\left(-x^2 + C\right)$$

$$\boxed{W = A\exp\left(-x^2\right)}$$

(Product Rule)

$$\overbrace{A\exp\left(-x^2\right)y'' - 2xA\exp\left(-x^2\right)y'}'$$

$$+ A\exp\left(-x^2\right)\lambda y = 0$$

$$\boxed{\frac{d}{dx}\left(A\exp\left(-x^2\right)y'\right) = -A\exp\left(-x^2\right)\lambda y}$$

c.) $x^2y'' + xy' + (x^2-\lambda^2)y = 0$

(Bessel ODE)

$$W = \frac{1}{P_0}\exp\left(\int \frac{P_1}{P_0}\,dx\right)$$

$P_0 = x^2$, $P_1 = x$

$$W = \frac{1}{x^2}\exp\left(\int \frac{1}{x}\,dx\right)$$

$$W = \frac{A}{x^2}(x) = \frac{A}{x}$$

$$\boxed{W = \frac{A}{x}}$$

$$\left(\frac{A}{x}\right)\left[x^2y'' + xy' + (x^2-\lambda^2)y\right] = 0\left(\frac{A}{x}\right)$$

$$Axy'' + Ay' + \frac{A(x^2-\lambda^2)y}{x} = 0$$

$$\boxed{\frac{d}{dx}\left(Axy'\right) = -\frac{A(x^2-\lambda^2)y}{x}}$$

Prob 3.)

Given:

* $y'' = -\lambda y$
* $y = A\exp(\gamma x) + B\exp(-\gamma x)$
* $-\lambda = \gamma^2$ and γ (Real eigenvalues)
* $y(0) = y(1) = 0$ (Dirchlet boundary conditions)
* $y'(0) = y'(1) = 0$ (Neumann b. conditions)

a) Dirchlet Conditions:

$y(0) = 0$:

$$0 = Ae^0 + Be^0$$

$$A = -B$$

$y(1) = 0$:

$$0 = Ae^{\gamma} + Be^{-\gamma}$$

$$0 = Ae^{\gamma} - Ae^{-\gamma}$$

$$0 = A(e^{\gamma} - e^{-\gamma})$$

- $0 = A\left(e^{\gamma} - e^{-\gamma}\right)$
- $0 = 2A\sinh(\gamma)$
- $\sinh(x) = -i\sin(ix)$
- $0 = -2Ai\sin(i\gamma)$
- $\sin(i\gamma) = 0$
- $i\gamma = n\pi$
- $\gamma = \frac{n\pi}{i}$

Using Dirichlet boundary conditions produces unreal eigenvalues, when we know that they are real. So we can't use Dirichlett B.C.'s.

b.)

- $y = Ae^{\gamma x} + Be^{-\gamma x}$
- $y'(0) = y'(1) = 0$
- $y' = A\gamma e^{\gamma x} - B\gamma e^{-\gamma x}$
- $0 = A\gamma e^{0} - B\gamma e^{0}$
- $A = B$
- $0 = A\gamma e^{\gamma} - B\gamma e^{-\gamma}$
- $0 = A\gamma(e^{\gamma} - e^{-\gamma})$
- $0 = e^{\gamma} - e^{-\gamma}$
- $0 = 2\sinh(\gamma)$
- $-i\sin(i\gamma) = 0$

$i\gamma = n\pi$

- Neumann conditions lead to unreal eigenvalues, so Neuman conditions won't work.

4.)

Givens:

* $u_n = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad (n = 1, 2, 3, \ldots,)$

Orthornormal & Complete over $(0, L)$

* $f(x) = \sum_n c_n u_n(x)$

* $c_n = \frac{\langle u_n | f \rangle}{\langle u_n | u_n \rangle} = \frac{\int_0^L u_n^* f(x)\,dx}{\int_0^L u_n^*(x)\, u_n(x)\,dx}$

a.) $f(x) = 1$

$$\int_0^L u_n^*(x)\, u_n(x)\,dx = \frac{2}{L}\int_0^L \sin^2\left(\frac{n\pi x}{L}\right)dx$$

$$\cos(2\theta) = 1 - 2\sin^2\theta$$
$$2\sin^2\theta = 1 - \cos(2\theta)$$
$$\sin^2\theta = (1 - \cos(2\theta))/2$$

$$\frac{2}{L}\int_0^L \frac{1}{2}\,dx - \frac{2}{L}\int_0^L \frac{1}{2}\cos\left(\frac{2n\pi x}{L}\right)dx$$

$$\frac{1}{L}\int_0^L dx - \frac{1}{L}\int_0^L \cos\left(\frac{2n\pi x}{L}\right)dx$$

$$\frac{1}{L}(x)\Big|_0^L - \frac{1}{L}\int_0^L \cos\left(\frac{2n\pi x}{L}\right)dx$$

$$u = \frac{2n\pi x}{L}$$

$$du = \frac{2n\pi}{L}\,dx$$

$$\frac{1}{L}(x)\Big|_0^L - \frac{1}{2n\pi}\int_0^{2n\pi} \cos(u)\,du$$

$$\frac{1}{L}(x)\Big|_0^L - \frac{1}{2n\pi}(\sin(u))\Big|_0^{2n\pi}$$

$$\frac{1}{L}(x)\Big|_0^L = 1$$

$$\langle u_n | u_n \rangle = 1$$

$$\langle u_n | f \rangle =$$

$$\sqrt{\frac{2}{L}} \int_0^L \sin\left(\frac{n\pi x}{L}\right) dx$$

$$u = \frac{n\pi x}{L}$$

$$du = \frac{n\pi}{L} dx$$

$$\sqrt{\frac{2}{L}} \left(\frac{L}{n\pi}\right) \int_0^{n\pi} \sin(u)\, du$$

$$\sqrt{\frac{2L}{n^2\pi^2}} \int_0^{n\pi} \sin(u)\, du$$

$$\sqrt{\frac{2L}{n^2\pi^2}} \left(-\cos(u)\right) \Big|_0^{n\pi}$$

$$\sqrt{\frac{2L}{n^2\pi^2}} \left(-\cos(n\pi) + \cos(0)\right)$$

$$= \sqrt{\frac{2L}{n^2\pi^2}} \left((-1)^{n+1} + 1\right)$$

$$f(x) = \begin{cases} \sum_{n=1,3,5,7,\ldots}^{\infty} \sqrt{\frac{16}{L n^2\pi^2}} \sin\left(\frac{n\pi x}{L}\right), & n \text{ is odd} \\ 0, & n \text{ is even} \end{cases}$$

b.)

- $f(x) = x$
- $u_n = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$, $n = 1, 2, 3, \ldots,$

$$c_n = \langle u_n(x) \mid f(x) \rangle$$

$$c_n = \sqrt{\frac{2}{L}} \int_0^L x \sin\left(\frac{n\pi x}{L}\right) dx$$

$u = x \qquad dv = \sin\left(\frac{n\pi x}{L}\right) dx$

$du = dx \qquad v = -\frac{L}{n\pi} \cos\left(\frac{n\pi x}{L}\right)$

$$\sqrt{\frac{2}{L}} \left[\frac{-xL}{n\pi} \Big|_0^L + \frac{L}{n\pi} \int_0^L \cos\left(\frac{n\pi x}{L}\right) dx \right]$$

$$\sqrt{\frac{2}{L}} \left[\frac{-L}{n\pi}(L) + \frac{L}{n\pi} \int_0^L \cos\left(\frac{n\pi x}{L}\right) dx \right]$$

$$\frac{L}{n\pi} \int_0^L \cos\left(\frac{n\pi x}{L}\right) dx$$

$u = \frac{n\pi x}{L}$

$$\frac{L^2}{n^2\pi^2} \int_0^{n\pi} \cos(u)\, du \qquad du = \frac{n\pi}{L} dx$$

$$\frac{L^2}{n^2\pi^2} \Big[\sin(u) \Big]_0^{n\pi}$$

$$c_n = -\sqrt{\frac{2}{L}} \left(\frac{L^2}{n\pi}\right) = -\sqrt{\frac{2L^3}{n^2\pi^2}}$$

$$C_n = -\sqrt{\frac{2L^3}{n^2\pi^2}}$$

$$f(x) = \sum_{n=0}^{\infty} -\sqrt{\frac{2L^3}{n^2\pi^2}}\sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$$

$$\boxed{f(x) = \sum_{n=0}^{\infty} -\sqrt{\frac{8L}{n^2\pi^2}}\sin\left(\frac{n\pi x}{L}\right)}$$

5.)

a.)

$$\int_a^b y_1\left[(py_2')' + qy_2\right]dx - \int_a^b y_2\left[(py_1')' + qy_1\right]dx$$

$$\int_a^b y_1\frac{d}{dx}(py_2')\,dx + \int_a^b y_1 q y_2\,dx - \int_a^b y_2\frac{d}{dx}(py_1')\,dx - \int_a^b y_2 q y_1\,dx$$

$$\int_a^b y_1\frac{d}{dx}(py_2')\,dx - \int_a^b y_2\frac{d}{dx}(py_1')\,dx$$

$$p\int_a^b y_1\frac{d}{dx}(y_2')\,dx - p\int_a^b y_2\frac{d}{dx}(y_1')\,dx$$

$$p\int_a^b y_1 y_2''\,dx - p\int_a^b y_2 y_1''\,dx$$

$u_1 = y_1 \qquad dv_1 = y_2''dx \qquad u_2 = y_2 \qquad dv_2 = y_1''dx$

$du_1 = y_1'dx \qquad v_1 = y_2' \qquad du_2 = y_2'dx \qquad v_2 = y_1'$

$$p\left[y_1y_2'\Big|_a^b - \int_a^b y_2'y_1'dx\right] - p\left[y_2y_1'\Big|_a^b - \int_a^b y_2'y_1'\,dx\right]$$

$$p\left(y_1y_2'\Big|_a^b - y_2y_1'\Big|_a^b\right) - p\int_a^b y_2'y_1'\,dx + p\int_a^b y_2'y_1'\,dx$$

$$\int_a^b y_1\left[(py_2')' + qy_2\right]dx - \int_a^b y_2\left[(py_1')' + qy_1\right]dx$$

$$= p\left(y_1y_2' - y_2y_1'\right)\Big|_a^b$$

b.) $\cdot\,(py')' + qy = f$

$\cdot\,(pG)' + qG = \delta(x-x')$

$$(py_2')' + qy_2 = f$$

$\cdot$ (multiply y_1) $\cdot$

$$y_1(py_2')' + y_1qy_2 = y_1f$$

$\cdot$ (multiply y_2) $\cdot$

$$y_2(py_1')' + y_1qy_2 = y_2f$$

• (subtract terms) •

$$y_1\left(py_2'\right)' - y_2\left(py_1'\right)' = f\left(y_1 - y_2\right)$$

$$y_1\left(py_2'\right)' - y_2\left(py_1'\right)' + fy_2 = fy_1$$

• Integration operator •

$y_1 = y$ and $y_2 = G(x,x')$

$$\int_a^b f(x)y(x)dx = p\int_a^b y_1\frac{d}{dx}\left(y_2'\right)dx - p\int_a^b y_2\frac{d}{dx}\left(y_1'\right)dx + \int_a^b f(x)G(x,x')\,dx$$

$$\int_a^b f(x)y(x)dx = \int_a^b f(x)G(x,x') + p\int_a^b y\frac{d^2G}{dx^2}dx - p\int_a^b G\frac{d^2y}{dx^2}dx$$

$$\int_a^b f(x)y(x)dx = \int_a^b f(x)G(x,x')dx + p\int_a^b yG''dx - p\int_a^b Gy''dx$$

$u_1 = y$, $dv_1 = G''dx$, $du_1 = y'dx$, $v_1 = G'$

$u_2 = G$, $dv_1 = y''dx$, $du_2 = G'dx$, $v_1 = y'$

$$\int_a^b f(x)\,y(x)\,dx = \int_a^b G(x,x')\,f(x)\,dx$$

$$+ p\left(y\frac{dG}{dx}\Big|_a^b - \int_a^b \frac{dy}{dx}\frac{dG}{dx}dx \right)$$

$$-p\left(G\frac{dy}{dx}\Big|_a^b - \int_a^b \frac{dy}{dx}\frac{dG}{dx}dx \right)$$

$$I = \int_a^b G(x,x')f(x)dx + py\frac{dG}{dx}\Big|_a^b - pG\frac{dy}{dx}\Big|_a^b$$

$$y(x') = \int_a^b f(x)y(x)\,dx + pG\frac{dy}{dx}\Big|_a^b = \int_a^b G(x,x')f(x)dx + py\frac{\partial G}{\partial x}\Big|_a^b$$